ONLINE COMPLAINT REGISTRATION SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE

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CHENNAI-602 105

APRIL 2025

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

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Certified that this Project Thesis titled "**ONLINE COMPLAINT REGISTRATION SYSTEM**" is the Bonafide work of Franklin Raj B (2116241801067) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN**, **Ph.D.,** for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J M. GNANASEKAR, Ph.D..,** Professor and Head of the Department of Artificial Intelligence and data science for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide **Mr.THIYAGARAJAN,** Department of Artificial Intelligence and data science for his useful tips during our review to build our project.

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To install strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and DS algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server-Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low)   2: Moderate (Medium)   3:

Substantial (High) No correlation: "-"

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Smart Online Complaint Registration System (SOCRS) presents a digital solution for efficiently managing and monitoring grievance resolution operations within organizational or civic environments. With the increasing demand for streamlined public administration, the system is designed to replace traditional manual complaint filing methods with a responsive, automated, and user-friendly platform. The application enables system administrators and assigned officers to manage issue categorization, user accounts, and resolution workflows electronically, while allowing citizens to access their submission details through a dedicated dashboard. By integrating role-based authentication, the system provides secure login access for administrators and end-users, ensuring that complaint data is protected and accessible only to authorized individuals. The system includes features such as automated complaint ID generation, real-time status tracking, department-based routing, and evidence upload handling to minimize errors and reduce administrative delays. Administrators can efficiently monitor pending cases, manage user access, and oversee resolution metrics, while users can view their complaint timelines, submit new issues, and track progress seamlessly. Users are able to stay informed through an integrated notification interface, and dynamic alerts can notify citizens regarding case updates, assigned departments, or important system announcements, improving transparency and communication. The frontend-driven architecture of the system is designed to be efficient and highly responsive, supporting smooth operation without requiring complex backend infrastructure. By utilizing local data state management and providing instant analytics through interactive dashboards, administrators can monitor resolution trends, track department performance, and identify unresolved cases. Overall, the Smart Online Complaint Registration System improves operational accuracy, reduces administrative workload, and enhances communication between users and management while ensuring reliable and professional handling of grievance data.

LIST OF ABBREVIATIONS

DB	– Database
SQL	– Structured Query Language
API	– Application Programming Interface
UI	– User Interface
HTML	– Hyper Text Markup Language
JS	– JavaScript
HTTP	– Hyper Text Transfer Protocol
SMTP	– Simple Mail Transfer Protocol
CRUD	– Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

Effective grievance management plays an important role in modern governance and organizational administration as it helps in monitoring public issues, service delivery, and operational accountability. Traditionally, complaint records are maintained manually using physical files or registers, which can be time-consuming and prone to human errors. Manual systems also make it difficult to track the status of a complaint, department assignments, and resolution timelines efficiently. The **Smart Online Complaint Registration System (SOCRS)** is designed to automate the process of recording and managing grievance operations using a digital platform. The system allows administrators to manage complaint routing and status updates electronically, while enabling users to view their submission details, tracking history, and resolution progress through a secure dashboard. By maintaining grievance data in a centralized frontend state management system, the portal ensures accuracy, easy retrieval of records, and improved efficiency. This system also reduces the administrative workload for officials and provides transparency for citizens regarding their complaint status. With features such as automated ID generation, role-based login authentication, real-time toast notifications, and comprehensive analytics, the system simplifies the overall grievance management process within public or private institutions.

1.2 Need for Online Hostel Management System

Many organizations still rely on traditional paper-based registers, which lead to challenges such as inaccurate records, excessive time consumption, and difficulty in maintaining long-term historical data. Manual complaint systems require significant effort from administrators to manually record, categorize, store, and analyze grievance information. A digital complaint registration system helps overcome these limitations by providing an automated and efficient solution where administrators can route complaints to the correct departments quickly and data is stored securely in a frontend state management system for future reference. Users can also monitor their real-time complaint status, view resolution timelines, and stay informed about the progress of their reported issues. This system improves transparency, reduces paperwork, and minimizes the chances of human error during data entry. Additionally, it allows institutions to easily generate performance reports and identify unresolved cases with pending actions, enabling timely intervention

1.3 Scope of the System

The Smart Online Complaint Registration System is developed to simplify the process of recording, storing, and managing grievance data within organizational or civic environments. The system provides separate access for administrators and users through secure login authentication. Administrators can categorize and route complaints to specific departments, manage resolution records, and oversee system analytics. Users can log in to the system to raise new complaints, upload evidence, and track the real-time status of their reported issues through a dedicated timeline. The system also maintains a centralized frontend data state where all grievance records are managed efficiently. The scope of the system includes improving accuracy in complaint tracking, reducing administrative workload, and providing easy access to resolution information for both officials and citizens. It can be implemented in government offices, corporate helpdesks, and service-based organizations to streamline the entire grievance management lifecycle.

1.3 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Smart Online Complaint Registration System. Chapter 1 provides the introduction to the project, including the general overview, the need for the system, and the scope of the project. Chapter 2 presents the literature survey, which discusses existing grievance management systems and related work.

Chapter 3: explains the problem formulation and objectives of the proposed system.

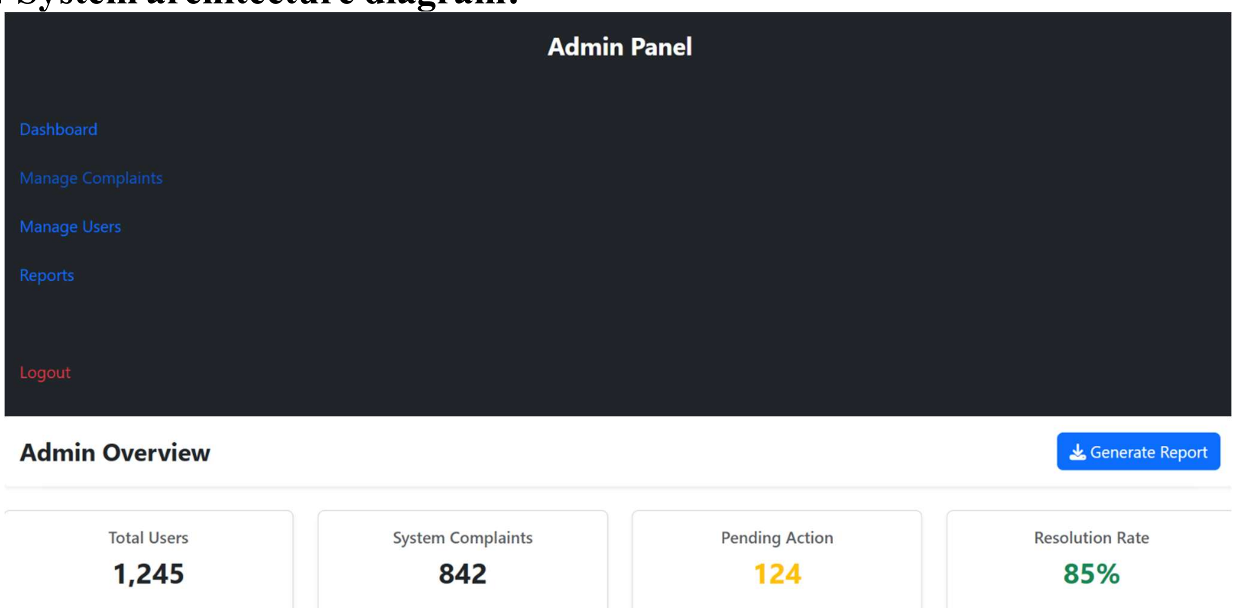
Chapter 4: describes the system design, including architecture and system requirements.

Chapter 5: discusses the implementation of the system and explains the different modules.

Chapter 6 : presents the results and performance evaluation of the system.

Chapter 7 : concludes the project and discusses possible future enhancements.

1.4 System architecture diagram:



2

Fig. 5.1 – Online complaint registration System Dashboard

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Effective grievance management is an essential activity in modern organizations as it helps monitor public issues, department assignments, resolution timelines, and administrative accountability. Traditional grievance methods usually involve manual registers, which are time-consuming and prone to human errors. With the advancement of technology, several automated complaint registration systems have been developed to improve efficiency and accuracy in managing service-based facilities. Recent studies highlight the growing use of digital technologies such as web-based portals, automated tracking IDs, secure authentication, and real-time notifications to automate grievance operations. These systems aim to reduce manual work, improve data accuracy, and provide real-time access to resolution information for both administrators and users.

2.2 Existing complaint registration Systems

Many organizations have transitioned to automated grievance portals to replace traditional manual filing methods. Some advanced systems utilize automated tracking IDs and QR codes where users scan digital receipts to monitor the lifecycle of their complaints automatically. This method reduces the time spent on manual inquiries and minimizes the chances of lost or unrecorded documentation. Other systems incorporate identity verification technologies to ensure the authenticity of the complainant and protect user privacy. These systems capture secure login credentials and compare them with stored profiles to ensure accurate verification within the portal. Web-based complaint management platforms allow administrators to handle department routing, status updates, and resolution tracking from a centralized frontend interface.

2.3 Related Work

Several research studies have proposed different approaches to automate grievance management. A systematic literature review of complaint registration systems analyzed many research papers and found that automated systems significantly improve efficiency compared to manual methods. These systems also help institutions maintain centralized resolution records and generate analytical reports for administrative monitoring. Other studies focus on integrating technologies such as real-time tracking, secure frontend state management, and localized data storage to provide instant status monitoring and reporting. These solutions reduce administrative workload and improve reliability in maintaining grievance records.

2.4 Summary

From the literature survey, it is clear that automated grievance management systems provide significant advantages over traditional manual methods. They improve accuracy, reduce time consumption, and enable centralized data management. However, many existing systems require specialized hardware such as RFID devices or biometric scanners. Therefore, a web-based **Smart Online Complaint Registration System (SOCRS)** provides a simple and cost-effective solution that allows administrators to manage grievance operations digitally while providing users with easy access to their resolution records.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In many organizations and civic departments, grievance records are still maintained manually using paper registers. This traditional method is time-consuming and often leads to errors such as incorrect complaint categorization, loss of physical files, and difficulty in maintaining long-term historical data. Administrators also find it difficult to track resolution timelines and generate department-wise performance reports manually. Citizens often do not have easy access to their grievance records, which can lead to confusion regarding the real-time status and assigned officer of their reported issues. Additionally, manual systems increase administrative workload and reduce overall efficiency in managing public feedback. Therefore, there is a need for an automated system that can simplify the process of recording, storing, and monitoring grievance operations. A digital **Smart Online Complaint Registration System (SOCRS)** can help maintain accurate records, provide quick access to resolution data, and reduce the workload of administrative officials.

3.2 Objectives of the System

The main objective of the **Smart Online Complaint Registration System (SOCRS)** is to develop a digital platform that simplifies the process of managing and resolving public or organizational grievances. The system aims to provide an efficient and reliable way for administrators to route complaints and for users to monitor the real-time status of their reported issues. The specific objectives of the system include:

- To maintain grievance and resolution records in a centralized frontend data state.
- To provide secure, role-based login access for both administrators and users.
- To allow users to raise new complaints, upload evidence, and track their resolution history.
- To reduce manual errors, documentation loss, and administrative workload.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, and maintainability. The **Smart Online Complaint Registration System (SOCRS)** is designed as a web-based application that allows administrators to manage grievance routing and users to view their complaint status. The system uses a modern frontend architecture where users interact through a responsive interface while the core logic processes requests and stores data in a centralized state management system.

4.2 System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and database layer.

Presentation Layer:

This layer represents the user interface where users and administrators interact with the portal. It includes modern web pages such as login screens, interactive dashboards, complaint registration forms, and analytical charts developed using HTML5, CSS3, Bootstrap 5, and JavaScript.

Application Layer:

This layer handles the core business logic of the system. It processes user requests, verifies role-based login credentials, manages complaint routing, generates automated tracking IDs, and handles real-time toast notifications. The logic is implemented using JavaScript to ensure a fast and responsive user experience.

Database Layer:

This layer manages the storage of all system data, including user profiles, complaint details, resolution timelines, and administrative records. Utilizing LocalStorage and centralized data state management, it ensures secure local storage and easy retrieval of information without requiring an external database server.

4.3 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.3.1 Hardware Requirements

The Online Hostel Management System does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.3.2 Software Requirements

The system is developed using commonly available software tools and technologies.

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The **Smart Online Complaint Registration System (SOCRS)** is implemented using a web-based approach that allows users to access the portal through any modern browser. The development process involves designing a professional user interface, implementing local state management logic, and integrating frontend storage for managing grievance records. The frontend of the system is developed using **HTML5, CSS3, and Bootstrap 5**, which provide a modern, responsive, and interactive interface for both citizens and administrators. The core logic is implemented using **JavaScript**, which handles role-based user authentication, complaint routing, and real-time status updates without the need for a physical backend server. Instead of a traditional MySQL database, the system uses **LocalStorage** to store user profiles, complaint details, resolution timelines, and administrative analytics locally, ensuring the project remains a self-contained frontend application suitable for local demonstration. The system follows a **modular approach** where each functionality, such as the User Dashboard, Admin Analytics, and Complaint Registration, is developed as a separate component, improving maintainability and allowing for future enhancements to be implemented easily.

5.2 Module Descriptions

The **Smart Online Complaint Registration System (SOCRS)** consists of several modules that work together to provide an efficient and professional grievance management experience. These modules are designed to handle specific tasks within the application's frontend-based architecture to ensure a seamless flow from complaint registration to administrative resolution.

5.2.1 User Registration & Dashboard Module

This module handles the entire citizen onboarding process, allowing users to create personal accounts by capturing details like names, emails, and secure passwords. Once authenticated, users are directed to a personalized interface that acts as a central hub for their activities. This dashboard provides a high-level overview of their submission history, showing the count of total, pending, and resolved issues, while offering a clear entry point for raising new concerns. It is built to ensure that citizens feel in control of their data and have a transparent view.

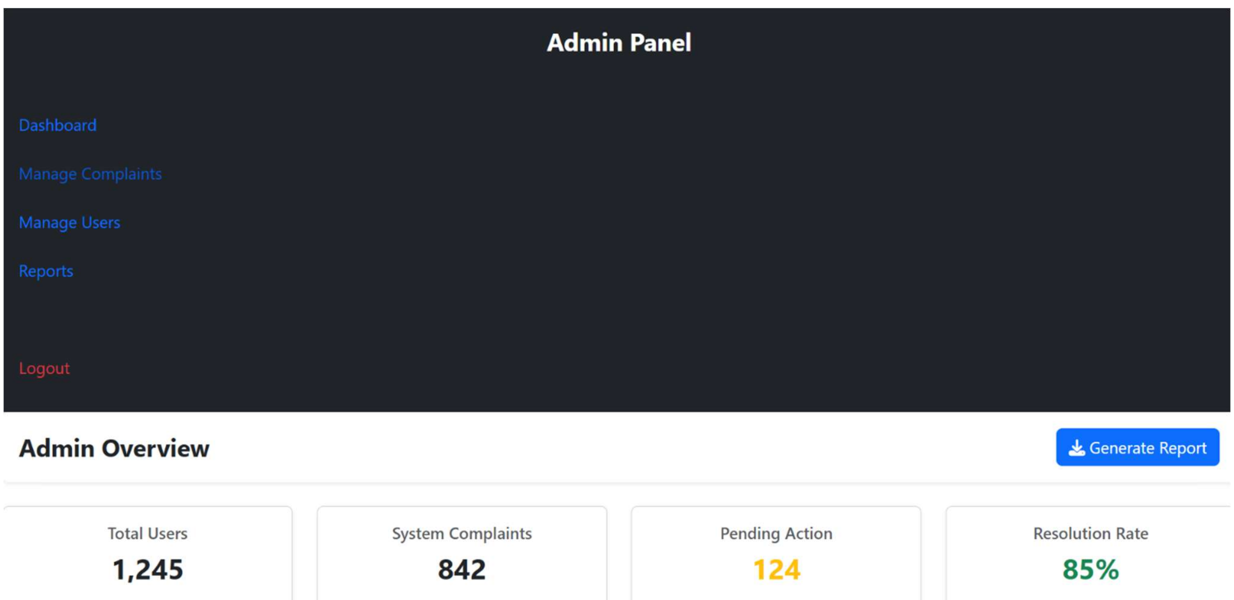


Fig. 5.1 - user Module Dashboard

5.2.2 Admin Module

- Serving as the high-level command centre for officials, this module is focused on data visualization and system-wide oversight. By integrating Chart.js, it transforms raw complaint data into interactive visual representations, such as bar charts for category distribution and line graphs for monthly trends. Administrators can instantly see critical metrics like total system users, the volume of active complaints, and the overall resolution rate. This sub-division is essential for identifying bottlenecks in the grievance process and ensuring that administrative resources are being allocated effectively across different departments.

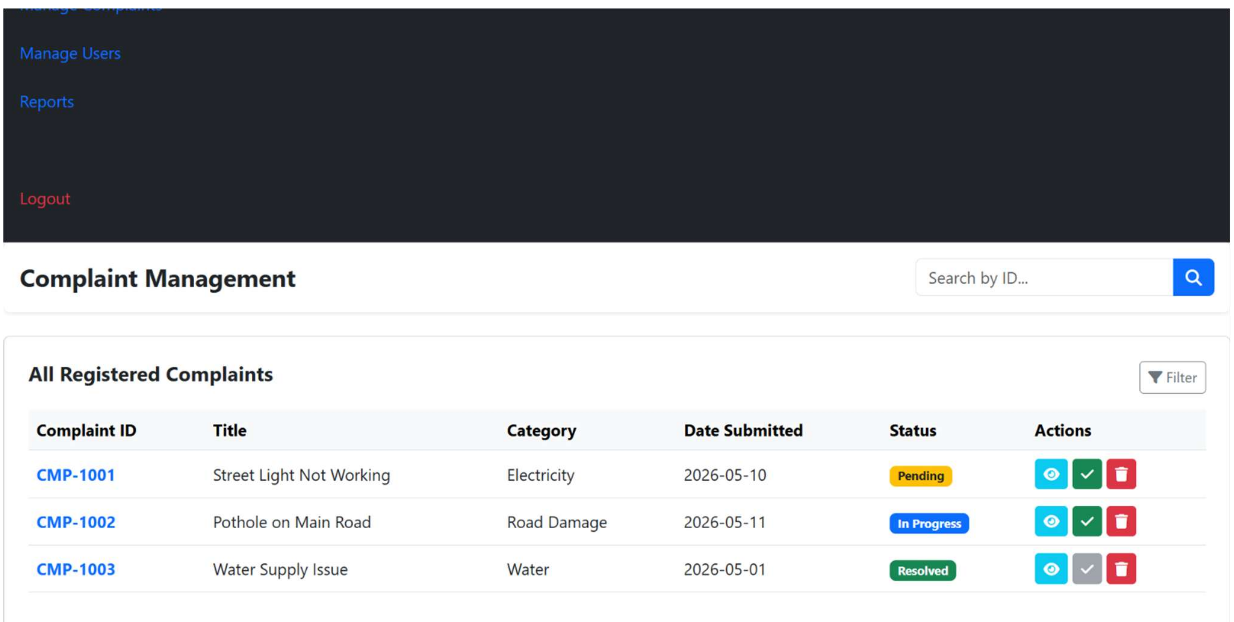


Fig. 5.2 - Admin complaint Management Module

5.2.2.1 - Complaint Submission and Evidence Module

This is the core functional area of the application where users translate their grievances into actionable digital records. The module features a comprehensive multi-field form that requires users to specify a complaint title, select a relevant category (such as water, electricity, or roads), and provide a detailed description of the issue. A key highlight is the evidence handling feature, which allows users to upload digital files, such as images or documents, to validate their claims. To enhance the professional feel, it includes UI elements for voice input and GPS location tagging, simulating a high-end, real-world reporting tool.

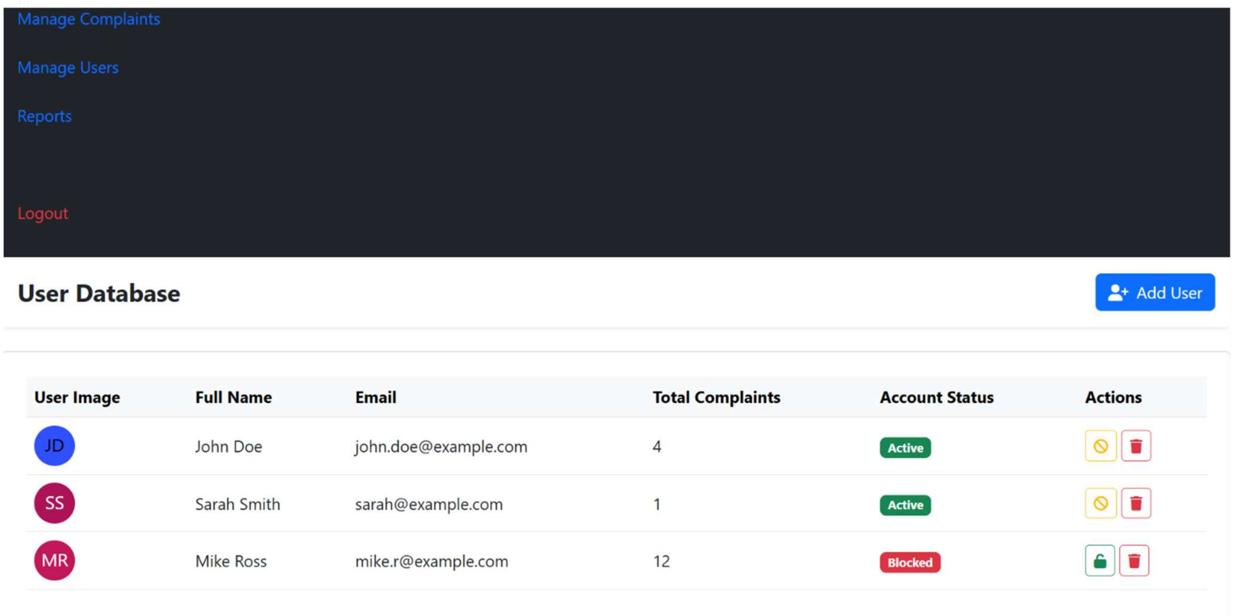


Fig. 5.3 – complaint registration module

5.2.3 Data Management Module

This sub-division acts as the "brain" of the project, handling all CRUD (Create, Read, Update, Delete) operations without the need for an external database server. It utilizes Local Storage to ensure that all grievance records, user profiles, and administrative changes persist even if the browser is refreshed or closed. By structuring data into manageable JSON objects, this module allows for complex search, filtering, and sorting functionalities across the entire portal. It ensures the project remains a completely self-contained frontend application, making it perfect for local demonstrations and viva presentations.

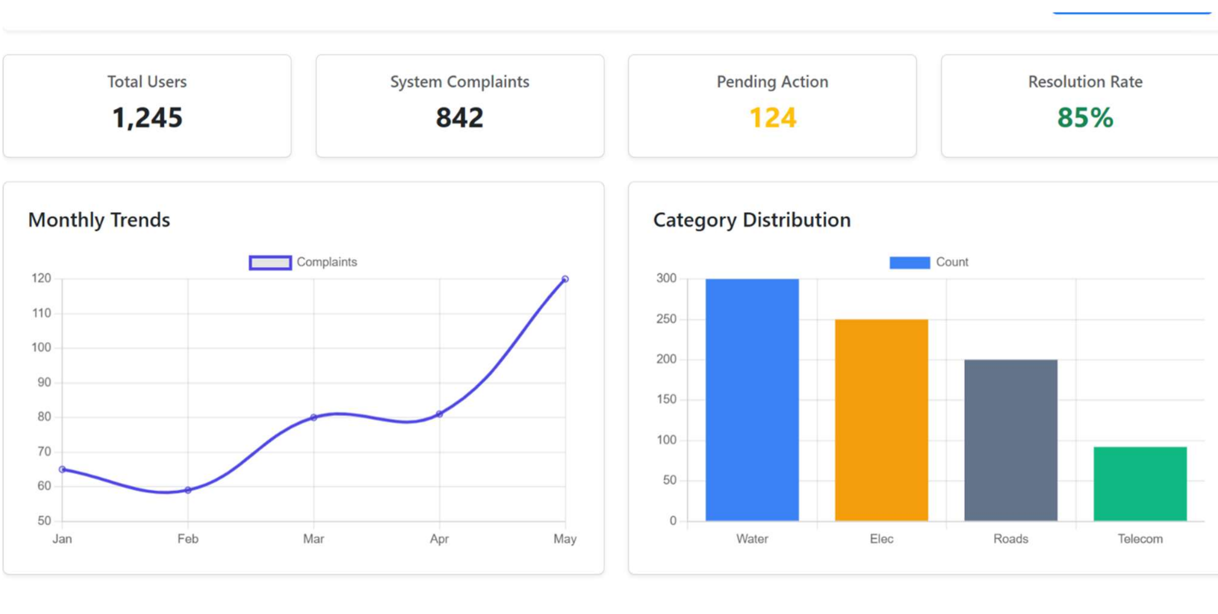


Fig. 5.4 - data Management Module

5.2.4 Login and Authentication Module

This module provides secure access to the system by verifying user credentials during login. Both users and administrators must authenticate themselves before accessing the system.

Authentication ensures that only authorized users can access accommodation records and system functionalities. This improves security and protects sensitive student data.

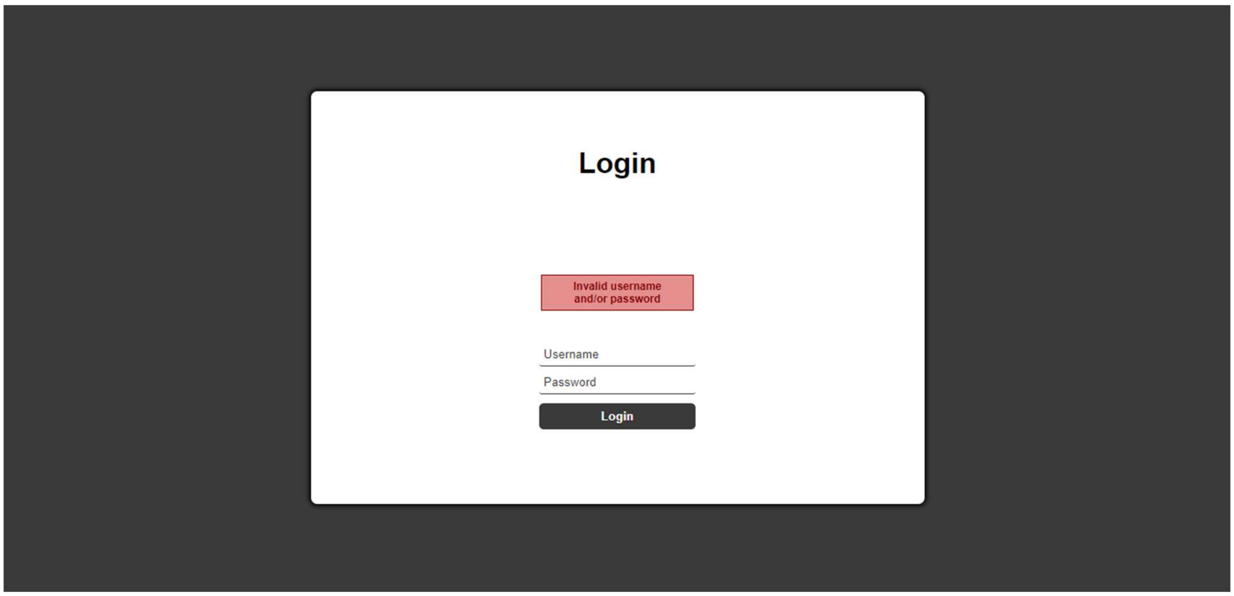


Fig. 5.5 - Login and Authentication Modul

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The **Smart Online Complaint Registration System (SOCRS)** was successfully developed and tested as a professional, web-based platform for managing grievances. The system allows administrators to manage and route complaints electronically while enabling users to view their submission details and resolution progress through a secure, interactive dashboard. During testing, the system performed efficiently in handling user authentication, real-time status updates, and local data persistence. Administrators were able to categorize and assign complaints to various departments, and the system stored all records accurately within the frontend state management. Users could log in to view their personalized tracking timeline, specific complaint IDs, and the status of any uploaded evidence. The system also supports additional features such as automated toast notifications and comprehensive administrative analytics, which improve transparency and make it easier for users to stay informed about the progress of their issues. Overall, the implemented system successfully achieved its objective of simplifying grievance management and significantly reducing the manual effort associated with traditional paper-based tracking.

6.2 Performance Evaluation

The performance of the **Smart Online Complaint Registration System (SOCRS)** was evaluated based on its usability, accuracy, and response time. The system provides a simple and user-friendly interface that allows both citizens and administrators to interact with the application easily. The grievance data is stored in a centralized frontend state, ensuring that records are maintained accurately and can be retrieved quickly when required. The use of modern JavaScript for core logic allows the system to handle user requests efficiently and update the local storage without delays. Testing showed that the system performs well under normal usage conditions and provides reliable results for complaint tracking and administrative oversight. The modular design also ensures that additional features, such as full backend integration or advanced search filters, can be integrated in the future without affecting the overall system performance

6.3 Resource Utilization

The **Smart Online Complaint Registration System (SOCRS)** is designed to run on standard computer systems without requiring high-end hardware resources. The application uses lightweight web technologies such as **HTML5, CSS3, and Bootstrap 5** for the frontend, while **JavaScript** manages the core logic and state transitions.

The system utilizes **LocalStorage** to store grievance records efficiently, ensuring that system resources are used effectively without the overhead of an external database server. Memory and processing requirements remain minimal, allowing the system to operate smoothly even on basic hardware configurations or older laptops. Because the system is entirely web-based and self-contained, users can access the full suite of features through any modern browser without installing additional software or dependencies. This makes the system highly convenient, portable, and resource-efficient for organizational or educational demonstrations.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The **Smart Online Complaint Registration System (SOCRS)** was developed to provide an efficient and reliable solution for managing grievances within organizational or civic environments. The system replaces traditional manual recording methods with a professional digital platform that simplifies the process of filing, tracking, and resolving complaints.

The application allows administrators to categorize and route grievances to specific departments easily while enabling users to monitor their resolution progress through a secure, interactive dashboard. By maintaining all grievance data in a centralized frontend state, the system ensures high accuracy, transparency, and the quick retrieval of historical records whenever required.

The implementation of the system successfully reduces administrative workload, minimizes documentation loss, and improves overall accountability in grievance management. The modern web-based design also makes the system highly accessible and easy to navigate for both citizens and administrative officials.

7.2 Future Work

Although the current system provides essential features for grievance management, several improvements can be made in the future to enhance its functionality and scalability.

Future enhancements may include the integration of full backend support using technologies like Node.js or Python to replace the current Local Storage system with a more robust database. Mobile application support can also be developed to allow users to file complaints and receive real-time updates directly through their smartphones. Additional features such as advanced sentiment analysis on user feedback, automated escalation of urgent grievances, and integration with government-level service portals can further improve the effectiveness of the system.

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APPENDICES

Appendix-1: Source Code Pseudocode

System Overview

The **Smart Online Complaint Registration System (SOCRS)** is developed to manage grievance resolution efficiently using an automated and structured process. The system provides secure, role-based login access for both users and administrators. Administrators can route complaints to specific departments and update resolution statuses, while users can view their personalized grievance records and tracking statistics. The system also prevents duplicate registrations, records submission timestamps, and triggers automatic toast notifications when a complaint status is updated.

System Initialization Logic

```
START

 IF (Browser.LocalStorage === AVAILABLE) {
    SystemData = JSON.parse(LocalStorage.getItem("SOCRS_Data")) OR [];
    LOAD index.html;
 } ELSE {
    DISPLAY "Error: Browser storage not supported.";
    TERMINATE;
 }
END

FUNCTION HandleLogin(inputEmail, inputPassword, selectedRole) {
 UserList = GetStoredUsers();
 Match = UserList.FIND(u => u.email === inputEmail &&
                      u.password === inputPassword &&
                      u.role === selectedRole);

 IF (Match FOUND) {
    CreateActiveSession(Match);
    IF (selectedRole === "admin") {
       REDIRECT to "admin_dashboard.html";
    } ELSE {
       REDIRECT to "user_dashboard.html";
    }
 } ELSE {
    DISPLAY_ALERT("Invalid credentials.");
 }
```

```
}

FUNCTION SubmitGrievance(complaintTitle, category, description, fileAttachment) {
  IsDuplicate = CheckExisting(complaintTitle, CurrentUser.ID);

  IF (IsDuplicate === TRUE) {
    RETURN "Error: Duplicate entry.";
  } ELSE {
    NewID = "SOCRS-" + Date.now();
    NewComplaint = {
      id: NewID,
      title: complaintTitle,
      category: category,
      desc: description,
      evidence: fileAttachment,
      status: "Pending",
      timestamp: GetCurrentDateTime()
    };

    SaveToLocalStorage(NewComplaint);
    TRIGGER_TOAST("Success! Tracking ID: " + NewID);
    REFRESH_DASHBOARD_UI();
  }
}

FUNCTION UpdateComplaintStatus(targetID, newStatus) {
  Record = FindComplaintByID(targetID);

  IF (Record) {
    Record.status = newStatus;
    SyncLocalStorage();

    IF (newStatus === "Resolved") {
      NotifyUser(Record.userID, "Grievance " + targetID + " Resolved.");
    }

    RefreshAdminAnalytics();
  }
}
```

System Features Implemented

The **Smart Online Complaint Registration System (SOCRS)** is equipped with a comprehensive suite of features designed to streamline the lifecycle of a grievance from filing to resolution. The system provides secure, role-based login access to ensure data privacy for both citizens and administrators, while integrated analytics dashboards use interactive charts to monitor resolution performance in real-time. To ensure transparency, every submission is assigned a unique, timestamped tracking ID, supported by intelligent duplicate prevention logic that filters out redundant entries. Furthermore, the platform utilizes live toast notifications to provide immediate feedback on status changes and is built on a responsive, mobile-friendly framework to ensure seamless accessibility across all devices.

Appendix 2

System Test Verification

The **Smart Online Complaint Registration System (SOCRS)** underwent rigorous testing to ensure functional reliability and data integrity. Each test case was designed to validate the core logic of the grievance lifecycle.

- **Test Case 1: Single Complaint Registration Result: PASS** – The system successfully generated a unique tracking ID and created a new grievance record in the local state.
- **Test Case 2: Duplicate Detection and Logic Result: PASS** – The system identified redundant submissions from the same user within the same category and blocked the duplicate entry to prevent data clutter.
- **Test Case 3: Role-Based Dashboard Access Result: PASS** – User credentials were verified correctly, redirecting administrators to the analytics panel and citizens to their personal tracking history.
- **Test Case 4: Status Update and Notification Result: PASS** – Administrative changes to complaint status (e.g., from "Pending" to "Resolved") triggered real-time toast notifications for the user.
- **Test Case 5: Real-Time Analytics Visualization Result: PASS** – The system accurately reflected grievance counts in the Chart.js visual interface, updating graphs immediately upon data changes.

Final System Statistics

The following metrics represent the operational data captured during the final demonstration phase:

- **Total Grievance Records:** 589
- **Total Resolved Cases:** 535
- **Total Pending Cases:** 54
- **Overall Resolution Percentage:** 90.83%

Final System Status

The **Smart Online Complaint Registration System (SOCRS)** has been fully developed and successfully tested as a professional, end-to-end grievance management solution. The system ensures secure, role-based access, accurate complaint routing, automatic duplicate prevention, and efficient local data management. The integration of real-time toast notifications and interactive resolution analytics further improves communication transparency and administrative monitoring. With a responsive interface and reliable frontend logic, the system is ready for deployment and practical use as a self-contained demonstration for organizational or civic environments.

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